Maldives Discovery

Design and Development of an Eight-Page Educational Travel Website

SD1004 | Introduction to Web Design

Student: [Full name] Student number: [CYX number] Batch: [Batch number]

Website URL: [Insert published URL after deployment]

Case Study | Section B1

Introduction

Maldives Discovery is an eight-page educational website developed for the SD1004 Introduction to Web Design assignment. Its purpose is to introduce readers to island geography, marine environments, culture, practical travel planning, accommodation categories and traditional food. The target audience is curious international visitors and readers who want a straightforward introduction rather than a booking platform.

The project uses HTML for semantic page structure, a shared external CSS stylesheet for presentation and JavaScript for interactive behaviour. Its visual identity combines deep teal, soft sand and pale aqua with original vector illustrations. The site was designed as a static, low-maintenance learning project that can be published through a static hosting service.

Objectives and design rationale

The primary objectives were to create a coherent eight-page site, make important content easy to locate, demonstrate responsive layout techniques, provide a form with client-side validation and include links to reliable external information. A secondary objective was to keep claims modest: the website is an independent educational demonstration, not an official travel authority or booking service.

A persistent navigation bar connects every page. The homepage introduces the subject and provides clear routes to island information and travel advice. The site uses short sections, descriptive headings, concise cards and illustrations to balance text and visuals. Deep teal provides contrast against white backgrounds, while warm sand panels distinguish supporting sections.

Methodology: planning and content

The project began by interpreting the assignment brief and identifying deliverables: eight interconnected HTML pages, one external stylesheet, JavaScript, a validated form, three relevant external links, original graphics, hosting and a case study. The subject was then divided into eight reader-focused pages: Home, Islands, Marine Life, Culture, Travel Tips, Where to Stay, Food and Contact.

Page content was written as concise introductory information. The site avoids dynamic prices, travel schedules and visa guarantees because these require current official verification. Links to Visit Maldives, Maldives Immigration and the UN Environment Programme guide readers toward appropriate sources.

Methodology: interface and implementation

Each page uses semantic HTML elements such as header, nav, main, section, article and footer. The shared assets/style.css file defines layout, typography, colour, spacing, card styles, form fields and responsive breakpoints. The same assets/script.js file is loaded by all pages and activates only relevant interactions when their controls are present.

Original SVG artwork includes a site logo and six thematic illustrations. SVG was chosen because it remains sharp at different display sizes and avoids large photographic downloads. The homepage uses a two-column hero on wide screens and a stacked layout on narrow screens. CSS grid supports three-column cards on desktop and fewer columns on smaller devices.

JavaScript handles the mobile navigation toggle, the island-category filter and the contact-form demonstration. Filter controls update visible cards and their aria-pressed state. The contact form checks name length, email format, message length, selected topic and acknowledgement before displaying a status message. It intentionally does not send or store personal data; adding a backend would be necessary for real enquiries.

Methodology: accessibility and quality checks

Accessibility considerations include descriptive page titles, a skip-to-content link, semantic headings, navigation labels, visible keyboard focus, descriptive link text, form labels, status announcements and a reduced-motion preference. The mobile menu exposes its expanded state with aria-expanded. The pages also use a viewport meta tag for responsive rendering.

Quality checks should cover opening all eight pages, verifying internal and external links, testing mobile menu operation, trying all island filters, and submitting both valid and invalid form entries. Browser testing should include keyboard navigation, narrow and

wide viewports and a review of console errors. These are proposed acceptance checks; results should only be recorded as completed after the student actually runs them.

Analysis and discussion

The consistent header and footer reduce the effort needed to move between topics. The subject-based navigation labels help users anticipate page content. Because all pages share one stylesheet, a visual adjustment can be made in one place rather than repeated in eight files. This supports consistency and reduces maintenance effort.

Original vector artwork gives the project a recognisable visual identity and keeps image files compact. The filter and FAQ components demonstrate useful interaction without requiring a complex framework. Client-side form validation gives immediate feedback, but it cannot replace server-side validation in a production system.

The project also has limitations. It does not provide live travel data, real booking functionality, analytics or a form-processing backend. It should not be presented as a fully operational tourism service. External sites and entry requirements may change; links and factual content should be reviewed before publication.

Information architecture and navigation

The information architecture follows a simple hub-and-spoke pattern. The homepage introduces the theme and provides direct links to major topics, while the same top-level navigation appears on every page. A visitor who arrives directly on the Marine Life page can still find Culture, Food or Contact without returning to the homepage. This supports users who enter through a search result or a shared link rather than the homepage.

The page labels were selected to be recognizable rather than decorative. For example, Travel Tips describes the practical planning information more clearly than a vague title such as Journey. Where to Stay indicates accommodation guidance without implying that reservations are available. Each page has a unique HTML title and a concise description to provide context in browser tabs and search results.

The footer repeats important navigation links and includes the Word and PDF versions of the case study. This gives readers an alternative route to key content and satisfies the brief requirement that case-study documents be linked from the website. The links are

relative, which makes them work in a local folder and on most static hosting platforms, provided the folder structure remains unchanged.

Visual design decisions

The design uses a restrained visual language rather than many unrelated effects. The primary teal suggests the surrounding ocean while remaining dark enough for important interface elements. Pale aqua separates introductory sections from the main white reading areas, and sand-colored sections add warmth. These colours are used consistently so that readers learn which elements are navigational, interactive or informational.

The headings use a heavier geometric sans-serif appearance while body copy uses a simpler sans-serif style with generous line spacing. This creates hierarchy without excessive font variation. The headline sizes scale with viewport width, and paragraphs have sufficient line height to support reading on smaller screens. Card borders, subtle background differences and consistent padding create grouping without visual clutter.

Each illustration was created as a self-contained vector graphic, including a custom square logo and themed coastline scenes. Vector assets can be displayed at different sizes without pixelation and are usually smaller than large photographs. Their use also avoids the need to obtain permission for third-party photographs. The illustrations are conceptual rather than documentary photographs, and the alternative text describes them accordingly.

Responsive implementation

The desktop navigation displays all eight page links in a horizontal row. At narrower widths the layout switches to a Menu button and a collapsible list. The button updates aria-expanded as the menu opens and closes. This approach keeps navigation accessible on a mobile screen without hiding important destinations behind tiny text.

The homepage hero and feature sections use two-column layouts when space permits. A media query changes them to a single column on smaller devices. Card grids similarly change from three columns to two and then one. This reduces horizontal scrolling and

prevents content from becoming too narrow to read. Buttons remain large enough to select with a finger, and form controls use the available width.

Responsive design should be verified at several widths rather than only by resizing a desktop browser once. Suggested checkpoints include a narrow phone, a tablet and a wide laptop display. Important checks include whether navigation is visible when expanded, whether card text wraps cleanly, whether images retain their proportions and whether form messages remain legible.

Form behaviour and validation

The Contact page provides a form with a name field, email field, topic selector, message area and acknowledgement checkbox. The layout places labels directly above their controls and uses asterisks to identify required inputs. The message field explains its minimum length. This arrangement is intended to make completion predictable and reduce unnecessary mistakes.

The form uses JavaScript to intercept submission and check the required information. A name shorter than two characters, malformed email, missing topic, message shorter than fifteen characters or unchecked acknowledgement produces an explanatory error message. Valid entries produce a success-style message explaining that the details passed validation but were not transmitted or stored. The form is then cleared.

The distinction between validation and delivery is important. Client-side checks can improve usability but cannot guarantee data integrity or security because browsers can be modified and scripts can be bypassed. A real contact service would need a server endpoint, server-side validation, abuse prevention, secure handling and an appropriate privacy notice. For this introductory assignment, the browser-only form demonstrates interaction without falsely claiming to send an email.

Testing strategy and evaluation

A basic functional test begins by opening index.html and following every main navigation item. Each destination should have a meaningful heading, the shared site identity and a working route to other pages. Next, each footer link should be followed, including the

case-study documents. A broken link at this stage may indicate a missing file or an incorrect relative path.

For the mobile menu, a tester should reduce the browser width until the Menu control appears, activate it, verify that all navigation items become visible and activate it again to close. The same interaction should be repeated with a keyboard using Tab and Enter or Space. The expanded state should change correctly, and focus should remain visible.

The island filter should be tested in three states: All, Local islands and Resort islands. The visible cards should match the selected category and the selected button should be visually distinct. A separate test should inspect whether the aria-pressed value matches the active filter. These checks confirm both visible behaviour and the accessible state communicated to assistive technologies.

Form tests should cover an empty submission, a short name, an invalid email, a missing topic, a short message, a missing acknowledgement and a complete valid entry. Each invalid case should show feedback without claiming success. A valid case should show the explicit demonstration notice. Since the project has no backend, testing should also confirm that no email or database transaction is implied by the interface.

Finally, after deployment, the published HTTPS address should be opened in more than one browser and at different viewport sizes. The student should record the test date, browser, device, result and any corrections made. A screenshot of the live homepage and the linked documents provides useful submission evidence. This report supplies a checklist rather than fabricated test results.

Hosting and maintenance plan

A static hosting service is appropriate because the project consists of HTML, CSS, JavaScript, SVG images and downloadable documents. The deployment should preserve the existing folder structure so that relative references such as assets/style.css and case-study/SD1004_Case_Study.pdf remain valid. The published root should contain index.html so that the hosting provider can serve it as the homepage.

After uploading the files, the student should open the live URL and inspect all eight pages. The browser developer tools can help identify missing resources or JavaScript errors. The student should then check the external resource links and verify that the PDF

and Word downloads open from the live site. Any changed domain or URL should be inserted into the case-study title page and presentation.

Future maintenance would include checking external links, reviewing potentially changing travel guidance and updating content as needed. Because all pages share the same style and script files, visual and interaction fixes can usually be made centrally. The website is intentionally limited to informational content, making static hosting suitable and keeping its operational requirements small.

Ethical and practical considerations

The project presents the Maldives as a real place with inhabited communities, not only a leisure destination. Its content therefore emphasizes respectful behaviour, environmental responsibility and the need to consult official information before making travel decisions. This avoids presenting broad generalizations as guarantees. The site also makes clear that its illustrations are artistic representations rather than photographs of specific locations.

The use of a non-operational contact form is disclosed prominently on the Contact page. A visitor should never be led to believe that a message has reached a person when the browser has only checked input fields. In a future version, any collection of personal details would require a clear purpose, secure handling and a suitable privacy statement. For the current demonstration, data remains in the browser and is not intentionally stored.

The site deliberately avoids invented testimonials, live booking prices, unverified transport times and claims of official affiliation. These decisions improve credibility and reduce the risk that visitors act on inaccurate information. The design also avoids aggressive animation and prioritizes readable text, visible controls and simple navigation. The result is a modest but coherent example of how technical choices and responsible communication can reinforce one another.

Conclusion

The project demonstrates how a small static website can combine useful information, consistent visual design, responsive CSS and purposeful JavaScript. The main design

lesson is that clear content organization and predictable navigation matter as much as visual effects. Reusable styles and restrained interaction made the eight-page site easier to maintain.

The website is ready as a local static demonstration. Publication, live link verification and final browser-based testing must be completed and documented before claiming that the hosting component of the assignment is fulfilled.

References

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United Nations Environment Programme. (n.d.). Home. https://www.unep.org/

Visit Maldives. (n.d.). Home. https://visitmaldives.com/

Mozilla. (n.d.). MDN Web Docs. https://developer.mozilla.org/

World Wide Web Consortium. (n.d.). Web Content Accessibility Guidelines (WCAG) 2.2. https://www.w3.org/TR/WCAG22/

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Appendix A: Website page map

- Home — index.html
- Islands — islands.html
- Marine Life — marine.html
- Culture — culture.html
- Travel Tips — travel.html
- Where to Stay — stays.html
- Food — food.html
- Contact — contact.html

Appendix B: Acceptance test record (complete after testing)

☐ All eight HTML pages open without missing assets

☐ Header and footer navigation works on every page

☐ Mobile navigation opens and closes using keyboard and pointer

☐ Island filter shows the correct card groups

☐ Contact form rejects invalid input and accepts valid input

☐ Three external reference links open successfully

☐ Published website works over HTTPS on desktop and mobile

☐ PDF and Word case-study links work on the hosted site

Test date: _____ Browser/device: _____ Notes: _____

Appendix C: Deployment record

Hosting provider: [To complete] Live URL: [To complete] Deployment date: [To complete]

Evidence: capture screenshots of the live home page, navigation, contact validation and live case-study links after publication.

Photography edition: source and testing note

The revised website displays genuine photographs of Maldivian coastlines and reef islands from Unsplash and Wikimedia Commons. The photo-credits.html page identifies source pages, photographers, and licensing information. The original vector logo remains an original graphic. The photos are loaded from external image hosts, so an internet connection is required; a local illustration is used as a fallback if an image cannot load. Before submission, verify every image in the published website and record the actual results in the testing section. Do not describe tests or hosting as completed until performed.